



ZEMITS DUOCRATUS

Diode Laser IPL System

Section C

Clinical Information

Diode Laser

808 Diode System

Diode Laser Hair Removal System is based on laser with a wavelength of 808nm long pulse laser, penetrates the epidermis to hair follicles, based on the principle of selective photo thermal, the fluency of the laser is preferentially absorbed by the melanin in the hair, damaging hair follicles, then the hair loses its regenerative capacity. Its unique continuous pulse output and sapphire contact cooling technology, can effectively protect the skin and to achieve painless, rapid and permanent hair removal at the same time.

Zemits DuoCratus Diode Laser Hair Removal Machine is a Class 4 laser product, can produce high power density laser beam.

Can produce high power density laser beam (Look into the laser beam can cause blindness, direct contact with the skin can cause burns and ignite flammable items), be careful when using, operate by the professionals to avoid the risk of injury. Please read the operating instructions carefully before using the product. If you have more questions about safety, operation and use, please contact your distributor.

Technical Principles

Laser selective photo thermal effect on hair removal play a decisive role. Selective photo thermal effect is the basic principle of semiconductor laser hair removal, which refers to the use of the target (hair follicle melanin and dermal papilla nutrient vascular hemoglobin) for a specific wavelength with good absorption, and each pulse treatment time equal to Or less than the target relaxation time (i.e., the target temperature is reduced by half the time required), so that photo thermal effect is confined to the target, thereby preventing the target to absorb laser energy, the

heat generated by its surrounding tissue Causing damage and achieve the purpose of effectively damaging hair follicles and hinder hair regeneration, and no scar formation, the role of the performance of the degenerative hair follicles, regenerative hair pigmentation and so on, and the growth of follicles due to their pigmentation, easy to absorb Specific wavelength of the laser to achieve the purpose of treatment, so semiconductor laser hair removal has the security and the permanent effect of hair removal. The semiconductor laser wavelength of 808nm light penetration of skin tissue depth of about 2.5mm, its scope of action can cover different parts and

depth of the hair follicles. Melanin is dispersed in hair follicle stromal cells and is transported to the hair shaft during hair growth. Hair follicle epithelium, dermal papilla and hair follicles are rich in melanin, these melanin selective absorption of laser energy, the heat generated instantaneous local high temperature, destroy hair follicles and hair shafts, hinder and stop the hair growth.

Light energy is easily absorbed by hair follicle melanin and dermal papilla nutrient vascular hemoglobin, rapid photo thermal effect, when the hair follicle temperature rises sharply to a certain value, the hair follicles are thermally expanded, the hair follicle melanocytes rupture and Out of the pores by steam, while the dermal papilla nutrition blood vessels due to the destruction of hemoglobin coagulation, double effect to achieve effective hair removal.

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Application:

Intended to remove unwanted body hair (e.g., armpit, leg/arm, back, chest and abdomen, bikini line, and face) for hirsutism patients.

It is intended to be used by a trained professional physician to emit pulsed light of a wavelength around 808 nm, delivered by a dedicated handpiece, to destroy hair follicles by heat, used in medical institute.

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If the patient has any of the following medical conditions, the treatment should not be performed, unless the patient presents a specific written permission from his physician*:

Absolute contraindications:

- Pacemaker / Defibrillator (internal / external).
- Metal plates, or pins in the treatment area.
- Implants in the treatment area (metal, silicone, bio-alcamid or any material).
- Skin Cancer (past / present).
- Other Cancer types (present).
- Precancerous lesions.
- Heart diseases / heart failure.
- Pregnancy - current & 3 months post.
- During Breast feeding.
- Topical use of Hydroquinone, acids, bleaching agents, Retin A, etc., at the treatment site during the preceding 14 days.
- Immune depressants consumption.
- Photosensitizing medications / herbs consumption.
- HIV.
- Other immunosuppressive conditions.
- Light-induced / heat-induced diseases (Lupus, Porphyria, Epilepsy, and others).
- Active Herpes Simplex infection.
- Skin diseases (past / present) – such as Psoriasis, Dermatitis, Vitiligo, and others, located at the treatment area.
- Active Rosacea and/or acne.
- Keloid scars.
- Fragile, dry or sensitive skin.
- Consumption of Retinoids (Roaccutane, Isotretinoin, and others) during the previous 6 months (topical and/or oral).
- Steroids consumption (Cortisone, for example).
- Bleedings, hemorrhages.
- Tanned skin – in the last 4 weeks for skin types I-III in the last 8 weeks for skin types >IV (sun / artificial tanning / tanning lotions).
- Tattoo / makeup tattoo in the treated area.
- Fertility treatments.

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Relative Contraindications

1. Any general none balanced health problems.
2. Non-skin Cancer (past).
3. Hormonal changes.
4. Hormonal imbalance.
5. Blood pressure problems.
6. Coagulation problems & anticoagulants / pro-coagulants consumption.
7. Liver and kidney diseases.
8. Arthritis.
9. Tendency to develop active Herpes Simplex infection (prophylactic can be considered with patient's physician).
11. Endocrine diseases (diabetes, polycystic ovaries for HR).
12. Menstrual cycle irregularities.
13. Antibiotics consumption during the 3 previous weeks.
14. Consumption of antidepressant / Antipsychotic medications.
15. Anemia.
16. Fractures.
17. Respiratory problems / diseases.
18. Epilepsy.
19. Connective and muscular tissue diseases.
20. All types of surgical treatments performed during the 2 previous months.
21. Deep / moderate chemical peel performed during the 3 previous months

Specific written permission should contain the physician's reference to the following details:

- The patient specific medical condition.
- The specific intended treatment
- The physician specific permission to perform the treatment related to the patient's medical condition.
- This permission should be documented in the patient's records.

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Safety

- Operation of the device must be performed only by certified operators.
- Never look directly at the light emitted from the hand-piece or at any reflecting surfaces during operation, even when protective eyewear is in use.
- Ascertain that the hand-piece is never directed at anything other than the treated area.
- Ascertain that all the operators, patients, and service personnel are wearing the appropriate eye protection wear, as supplied with your hand-piece.
- Use professional eyewear only.
- We highly recommend removing/covering all potentially-reflecting surfaces from the treatment room.
- Mirrors shall not be placed in the room.
- Ascertain that skin area, other than the treatment site, are properly protected.
- Never operate the device when the plastic hand-piece cover is open.
- Never press the pulse trigger button of the hand-piece before the hand-piece spot is entirely located on the treatment spot.
- Verify that any flammable and explosive liquids (such as ethanol and ether) and gasses are protected from the light route.
- Laser Radiation Warning Sign must be placed at the entrance to the treatment room.
- Only certified technicians are allowed to disassemble the hand-piece.
- Always set the system to standby mode when treatment is paused; it prevents undesired light emission.
- Ascertain that the hand-piece spot covers only and fully the skin area intended for treatment.
- Treatment of minors is subjected to Parent / Guardian and physician confirmations and may be subjected to other additional legislative measures. Refer to your local authorities regarding the relevant legislation

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Operating the System

Preparation

Before treatment, please confirm the below information

- 1) Check all the accessories of the system are connected well
- 2) Make sure the person wears the protection goggles or eye patch in the treatment zone
- 3) Make sure the emergency button is not pressed

Skin test treatment

(The following is the Skin test for short)

Skin test is important before treatment, details as below:

- 1) Select a representative site for experimental treatment. If hair removal treatment is performed on the whole face, it should be chosen not to affect the aesthetic position of the face, such as the ear root. If it is the skin of the chest, neck, or other part of the body, you should choose other areas that do not affect your visual appearance. Also note that the selected skin should not have any symptoms of injury or infection.
- 2) Before skin test, shave the excess hair with a razor and clean the skin surface. Apply a thin layer of transparent gel in the area of the skin test. The thickness is about 2mm to 3mm. Hold the laser treatment head and touch the skin vertically, and apply pressure to the hair growth direction for 2-3 times. Wait for 5-10 minutes to wipe the gel and observe the test results
 - If there is erythema around the hair follicle, no epidermal response is the best effect. The following treatment can be set according to the same energy and pulse type.
 - If there is no visible change in the skin of the skin test, the energy can be increased by 1-2j/cm² appropriately.
 - If the patient feels a severe pain or a blister in the area of the skin test, then the energy may be reduced due to high energy or excessive repetitive treatment.

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Operating the System

Please note:

- 1) The patient and the operator should wear protective glasses during the treatment, and the patient should stay in bed;
- 2) The treatment area should start from small and gradually expand;
- 3) Remove hair from the treatment area and clean the skin before treatment. Be careful not to damage the skin and ensure that the skin is not infected;
- 4) the outlet should be close to the skin and properly pressurized;
- 5) The treatment should follow the hair growth direction in order to slide the treatment;
- 6) Depending on whether the treatment skin is sensitive to determine whether anesthesia is required during treatment;
- 7) Keep in touch with the patient during the treatment, consult the patient feeling and adjust the treatment parameters reasonably;
- 8) Patients with darker skin and thicker hair should choose low energy density. Patients with lighter skin and thinner hair should have higher energy density;
- 9) Wash the treatment head with wet gauze at any time and keep the treatment head clean;
- 10) After treatment, the instrument must be closed to keep the instrument cool.

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The interface has a numerical selection of energy density, pulse width, skin type selection and frequency, which can be adjusted by corresponding buttons.

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Energy density:

single pulse energy density, based on skin type and frequency selection, adaptive adjustment range automatically, unit is J/cm². The adjustable total range is 1-120J/cm².

Pulse width:

pulse duration, according to the skin type and frequency select the adjustment range of the adaptive pulse width, the unit is ms.

Skin type:

the degree of pigment of the skin, there are 6 types of adjustable range.(I, II, III, IV, V, VI).

Frequency:

laser pulse output frequency, adjustable range is 1-10Hz.

Temperature:

real-time display of current treatment head temperature, unit is °C.

Total Shots:

shows that when the laser output count is treated.

Total energy:

Showing the current treatment laser output energy.

Ready:

press the "Ready" button and the system is ready to control the laser output. In this state, in addition to the shutdown button and standby button, other buttons are invalid.

Standby:

press the "standby" button, the system is in standby mode, no laser output.

Note:

After you click the standby button, if you do not click the "Ready" button within five minutes, the system will enter a low-power state and close the cooling system.

Warning:

A skin test must be performed before treatment; the increase in energy must be from low to high; the skin's optimum energy is obtained during the skin test, and then a large area of treatment is performed. When preparing for the state, it is recommended that the temperature of the applicator's head reach about 4 degrees, and then start to output the laser.

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Location selection: There are 7 optional parts, namely: face, chest and abdomen, armpit, back, bikini, arms, legs;

Skin type: The degree of pigmentation of the skin is divided into 1-6 files, and the default skin is I.

Energy density: It is divided into 1-6 files, the energy density gradually increases from 1 to 6, the default is 1 file.

Operation process: According to the selected part, skin type, and energy density, to confirm the total energy of this time out of the light. When the energy of the light reaches the total energy, you can stop the treatment.

Recommended amounts of shots per treated area:

Armpits: 200-300 (100-150 per each armpit)

Bikini: 450-650

Full bikini: 600-800

Full Legs: 5,000-6,000 (2,500 - 3,000 per each leg)

Calves: 3000 (1,500 per each calf)

Full Arms: 1200-2,000 (600 - 1,000 per each arm)

Forearms: 900-1,400 (450 - 700 per each forearm)

Upper lip: 60-100

Chin: 50-250

Areolas: 100-200 (50 - 100 per each areola)



Note:

After you click the standby button, if you do not click the "Ready" button within 1 minute, the system will enter a low-power state and close the cooling system.

Warning:

A skin test must be performed before treatment; the increase in energy must be from low to high; the skin's optimum energy is obtained during the skin test, and then a large area of treatment is performed. When preparing for the state, it is recommended that the temperature of the applicator's head reach about 4 degrees, and then start to output the laser.

Pre-Treatment Session

Whenever treatments are considered, a Pre-Treatment session must be scheduled first. The purpose of this session is to provide the patient with the full information regarding the treatment, record the patient baseline status and set the treatment parameter by performing a test session.

Consultation

Consultations must include comprehensive, detailed, professional description and explanation of the future treatment sessions.

Whenever treatments are considered, consultation meeting must be scheduled first.

Consultations must include comprehensive, detailed, professional description and explanation of the future treatment sessions.

The patient must be informed of any potential risks the treatment might pose, as well as of all benefits and costs. The patient must also be informed of the planned progress stages and duration of the treatment.

During consultation, the attending physician (or an authorized staff member) should:

- Obtain a detailed medical history of the patient with previous treatment modalities, and check the patient's condition for treatment suitability.
- Medical history and consent forms must be completely filled and signed by the patient.
- It is highly recommended that the consultant will review the forms with the patient, while clarifying accuracy of the answers and providing any relevant explanations, if necessary.
- Discuss the reasons why the patient is seeking treatment and understand patient expectations. Realistic expectations matching is one of the keystones in achieving optimal treatment satisfaction and perceived success.
- Fully detail and explain treatment conditions, indications, contraindications, as well as possible side effects, complications and realistic expectations.
- Thorough examination of the skin and body must be conducted by the professional consultant.
- Patient's skin type must be classified and recorded in accordance with Fitzpatrick skin classification.
- Patient's hair type and color must be classified and recorded.
- The treatment-intended area should be photographed
- All data must be recorded in the patient's records.

Diode Laser Treatment Protocol

Hair Removal Treatment Protocol

1. Clean and disinfect your working area.
2. Wear disposable gloves and keep them on during the entire procedure. Ascertain that protective eyewear is within reach.
3. Carefully Shave the intended treatment skin area. Pre-treatment: hair can not be waxed before Diode Laser hair removal.
4. Thoroughly clean any creams, perfumes, or any other substances that may have contacted the skin off the treated skin area. Dry the skin thoroughly.
5. Photograph the treated area.
6. Clean the intended treatment area.
7. Use a white pencil, intended for skin application, to mark the borders of the area you wish to treat. You can mark and divide the area into smaller regions, for the convenience of treatment performance.
8. Apply an even layer of conductive gel to the treated skin area.
9. Set the desired treatment parameters.
10. Set the Cooling on High.
11. Select the appropriate patient eye wear supplied with your device. Verify it is properly worn.
12. *Wipe the frost layer accumulated on hand-piece spot using a cloth and ethanol 70% prior to contacting the skin.
13. Wear the supplied operator protective eyeglasses, suitable for operating the device. Keep them on during the entire procedure.
14. Press READY button on the screen
15. Position the hand-piece on the skin and verify that the hand-piece spot fully covers the skin. Apply mild pressure to assure full contact.
16. Trigger a pulse by pressing on the button on the handpiece.
17. Begin the procedure: perform quick linear or circular motions with the laser tip, and repeat this technique over the treated zone until the needed total count of shots is performed.
18. Add gel to the next skin area (ensure you have constant contact with the gel), then continue the same technique on the other areas.
19. The patient is expected to report a slight stinging sensation.

Follow-up & Consecutive Treatments

Hair Removal Follow-up & Consecutive Treatments

- o Diode HR treatment intervals should be between 4 weeks.
- o Re-evaluation of these intervals should be made after performing 3 or 4 treatment sessions, since hair growth gradually slows.
- o Perform treatments ONLY when there is evident hair growth.
- o Multiple treatments are needed—6, 8, or 10, depending on the treated area

Always perform a skin test on the intended treatment area or on similar sun exposed areas prior to commencing treatment, in accordance with the recommended parameters.

Do not treat hair growing at tattoos or pigmented lesions sites. Ascertain that pulses are not performed within a safety distance of 1.5 cm from the tattoo / lesion borders.

For safety and efficiency considerations:

Since optimal light absorption occurs when the surrounding area color is much lighter than the hair shaft / follicle color, treatment of white hair and hair that is lighter than the surrounding skin is not allowed. Treatment applied under these conditions may result in a burn.

Possible Side-Effects

- IPL and Diode System can cause a certain degree of undesired tissue damage or create the following side effects:
- Discomfort
- Irritation, itching, or a mild burning sensation (similar to sunburn) may occur within 48 hours after the treatment.
- Redness and swelling may occur within 2-24 hours after the treatment.
- Blisters, epidermal erosions, or hyper pigmentation may develop and remain visible for several days following the treatment.
- Change of color (Pigmentation) – A change in pigmentation may occur in the treated area. Most cases of hypo/hyper pigmentation occur in people with darker skin or when the treated area was exposed to sunlight before or after treatment. For some patients, hyper-pigmentation occurs despite their being protected from the sun before/after the treatment. The hypo/hyper pigmentation usually fade after three to six months; however, in rare cases (mainly hypo-pigmentation); the pigment change may be permanent.
- Fragile Skin – The skin at or near the treatment site may become fragile. Makeup should be avoided, and the area should not be scratched or rubbed.
- Damage to Natural Skin Texture – In some cases, there may be formation of a blister/crust, which may take as long as five to ten days to heal.
- Swelling and Bruising – Immediately following treatment, the skin may swell temporarily, especially in the nose and cheek areas. This swelling usually heals within hours, but may last up to seven days.
- Purpura (purple bruising) may appear on the treated area, and lasts from five to fifteen days. It can take up to three months for the entire bruise to disappear/discolor.
- Scarring – There is a slight chance of enlarged hypertrophic scarring on darker skin. To reduce the chance of such scarring, it is important to carefully follow the post- treatment stage instructions.

OUR CONTACT INFO

Have a question? Contact us.

Our Business Hours: :

09:00 AM - 6:00 PM, Monday to Friday (Eastern Standard Time)

Place an Order Online 7 Days a Week 24 Hours a Day

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